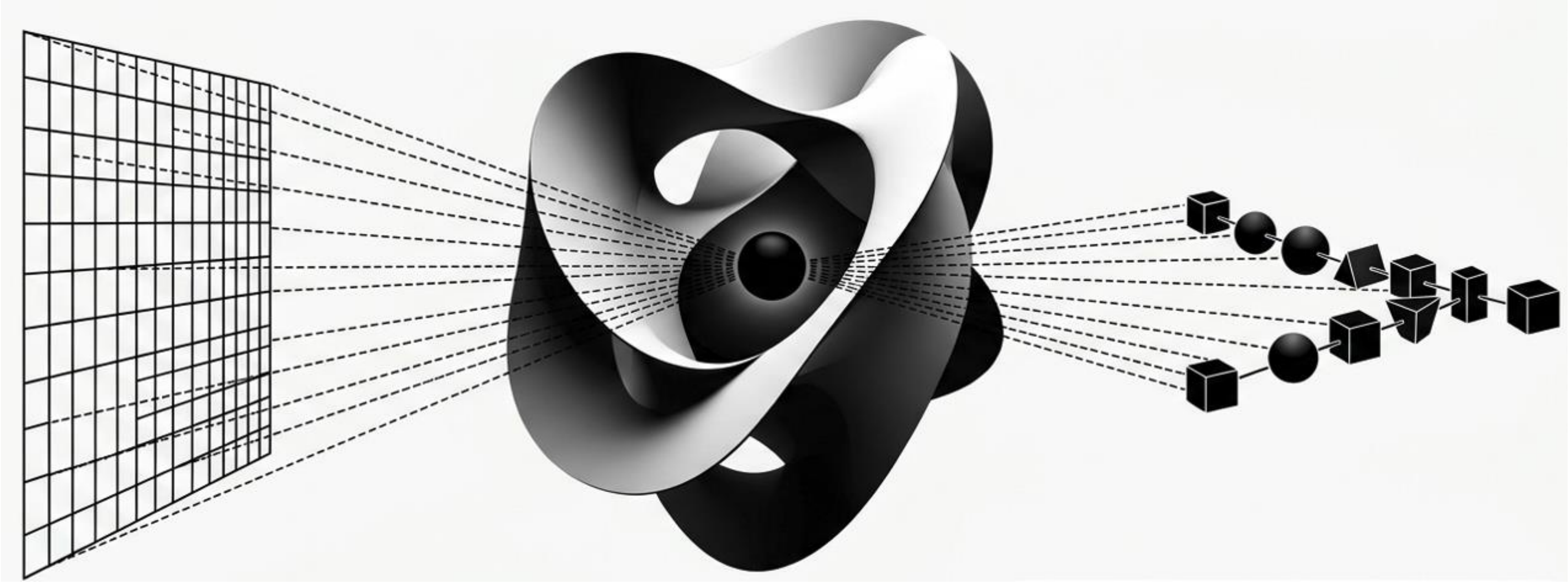
Knowledge mapping to the same reality
- multimodal similarity learning

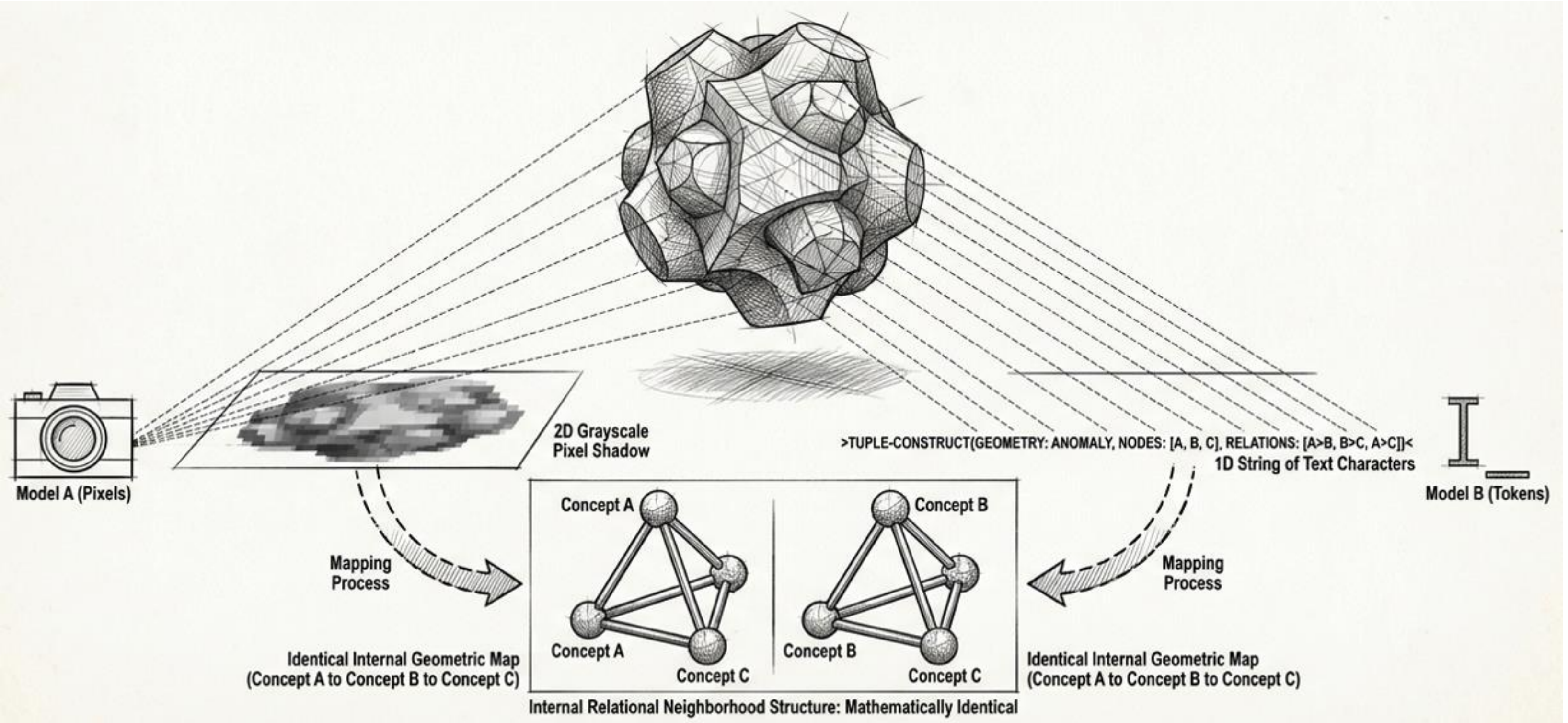


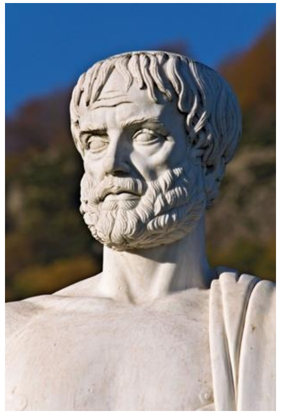
Synthesis of vision, language, and mathematical topology of medical knowledge

Stanley Liang, PhD

The Aristotelian Representation Hypothesis (ARH)

Different, sufficiently capable models trained on the same underly reality will naturally learn similar relational neighborhood structures, even if their internal architectures and input modalities differ entirely





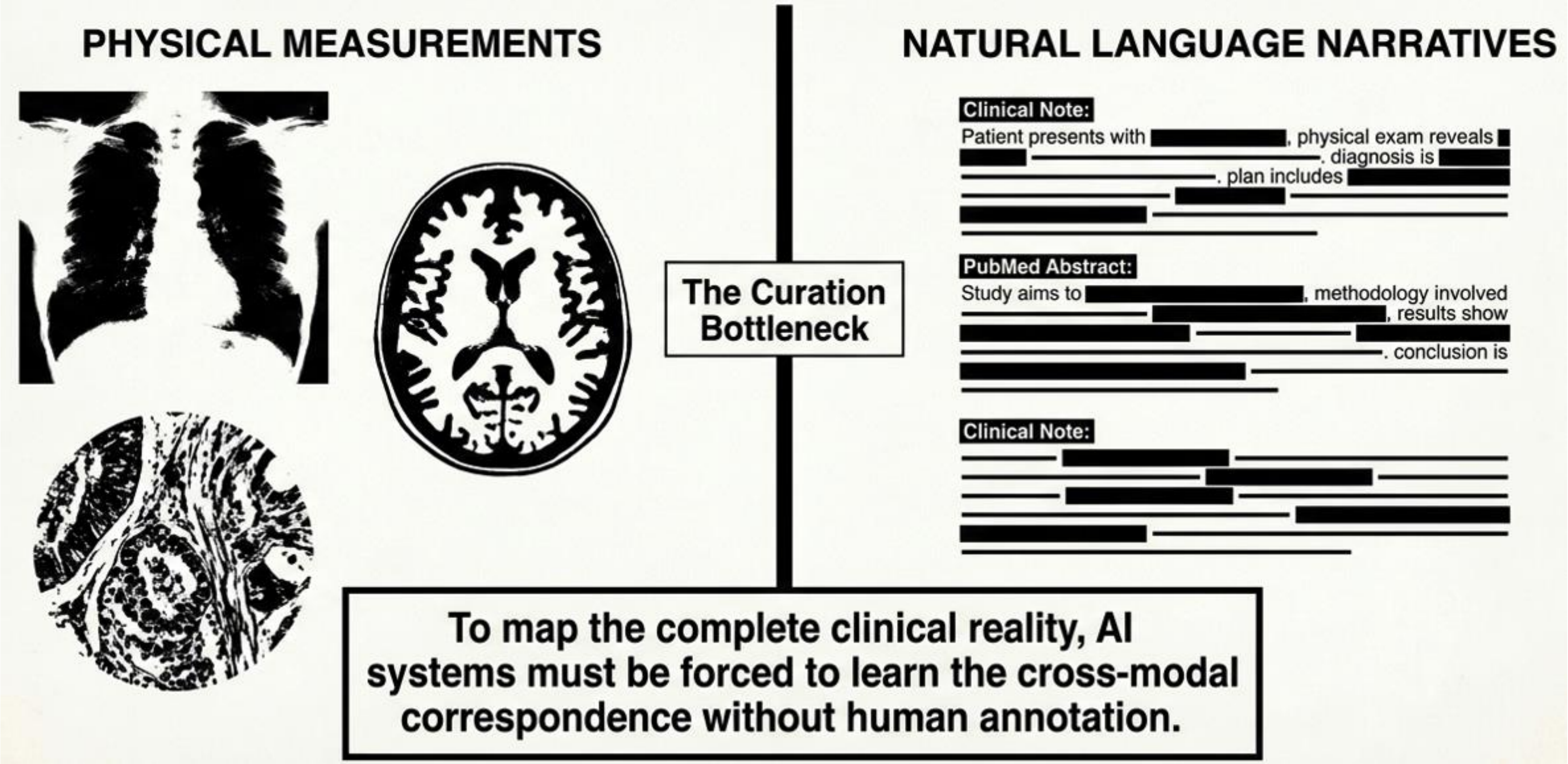
Aristotle

Modality is just a perspective, the underly truth is the same.

A modality is simply a different format of data that, when processed by a sufficiently capable model, will still map to the same shared "social circles" of information

The Biomedical Data Dichotomy

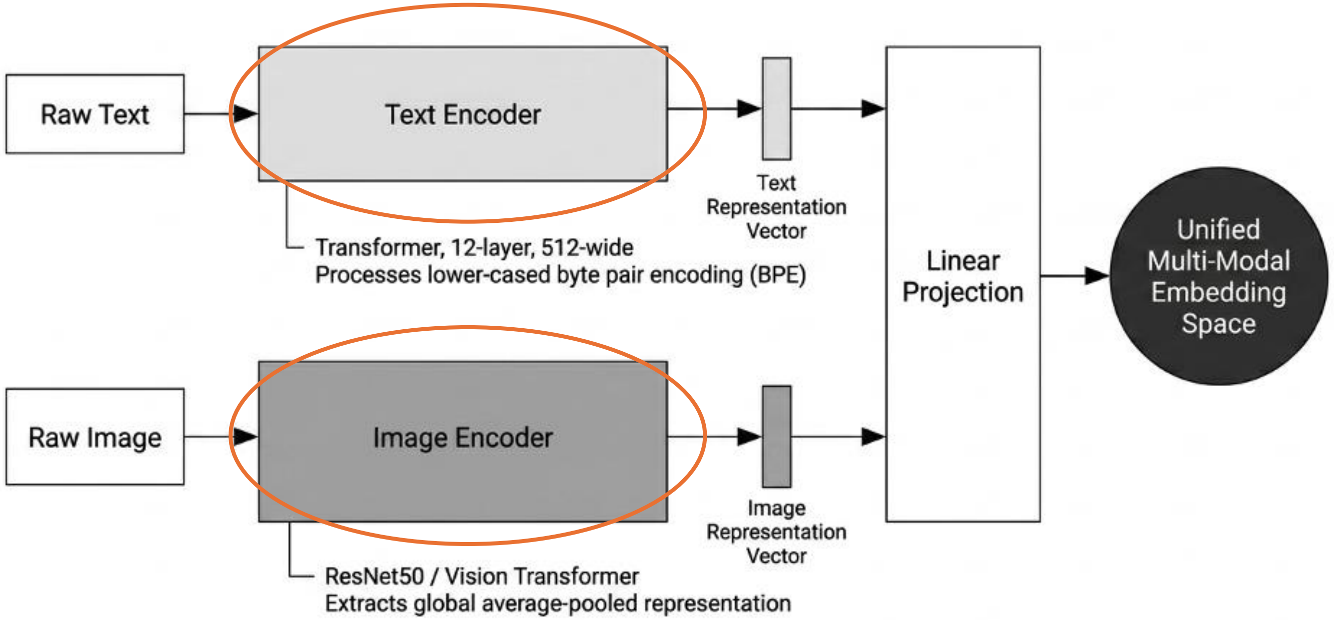
Medical knowledge exists simultaneously in images and text narratives. However, manual curation to link these modalities cannot scale.



- Manually annotated and structured data requires expensive expert labor
- Multimodal models can uncover latent predictive signals without relying on annotated data

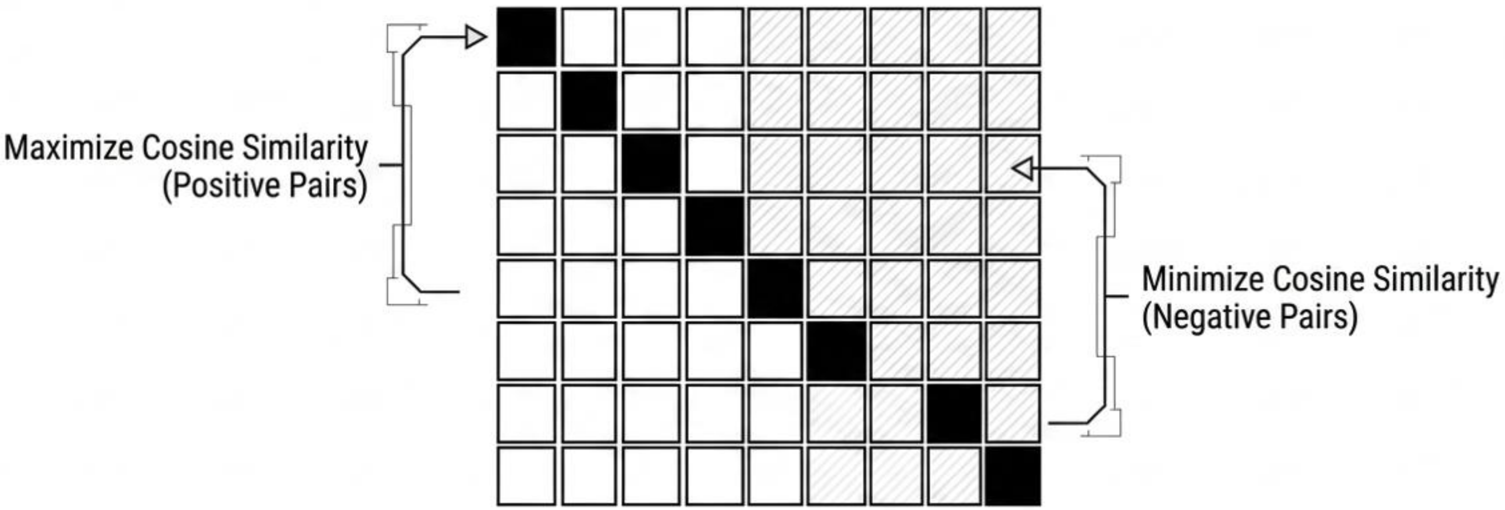
Solution – Multimodal Similarity Learning

Contrastive Language-Image Pre-training (CLIP) simplifies the task by predicting the entire text pairs with which image across a batch

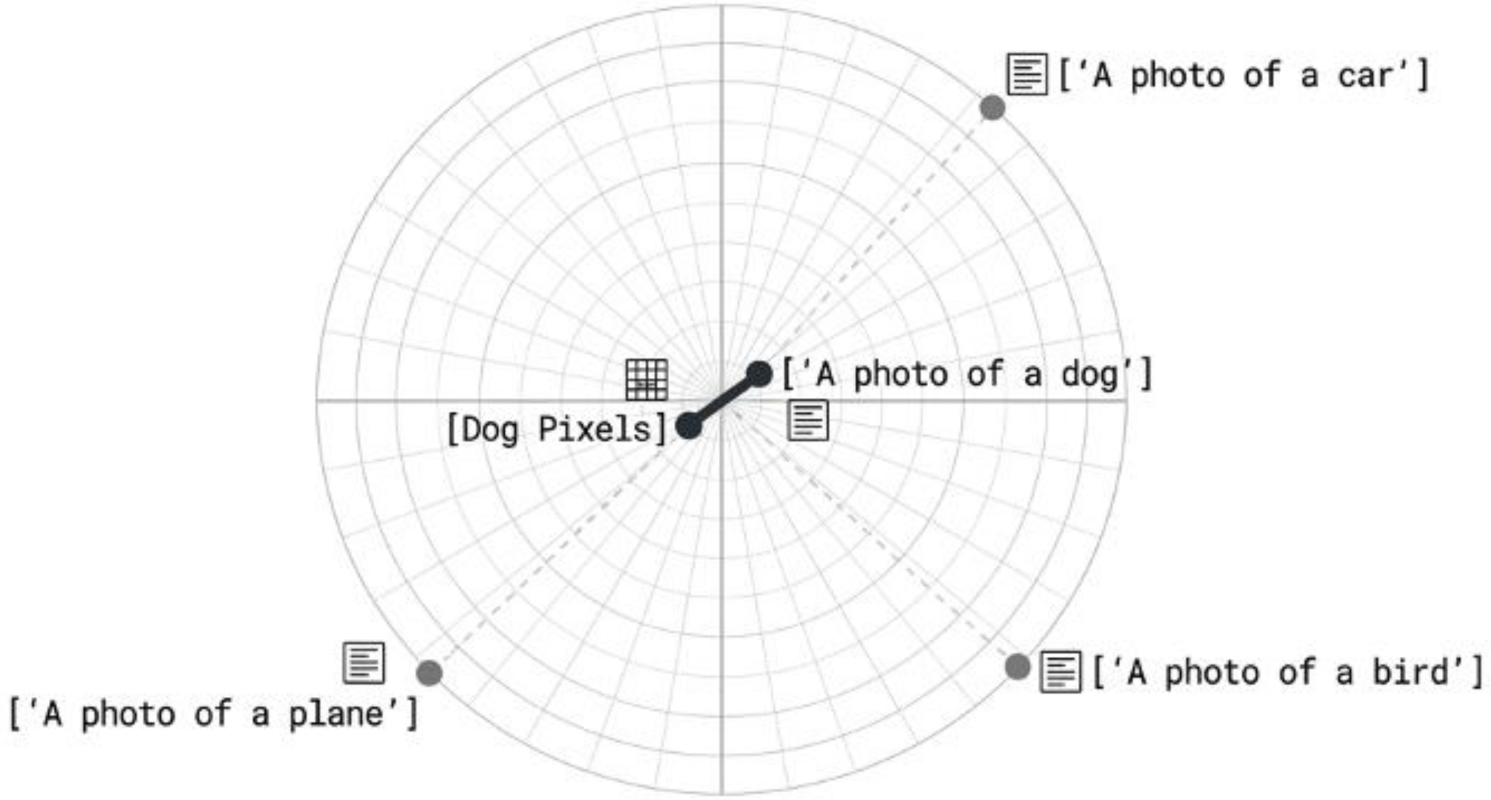


The symmetric cross-entropy loss (Information Noise-Contrastive Estimation, **InfoNCE**) forces the text and image encoders to project their distinct inputs into a single, shared multi-modal embedding space.

Core architecture: Dual Encoders

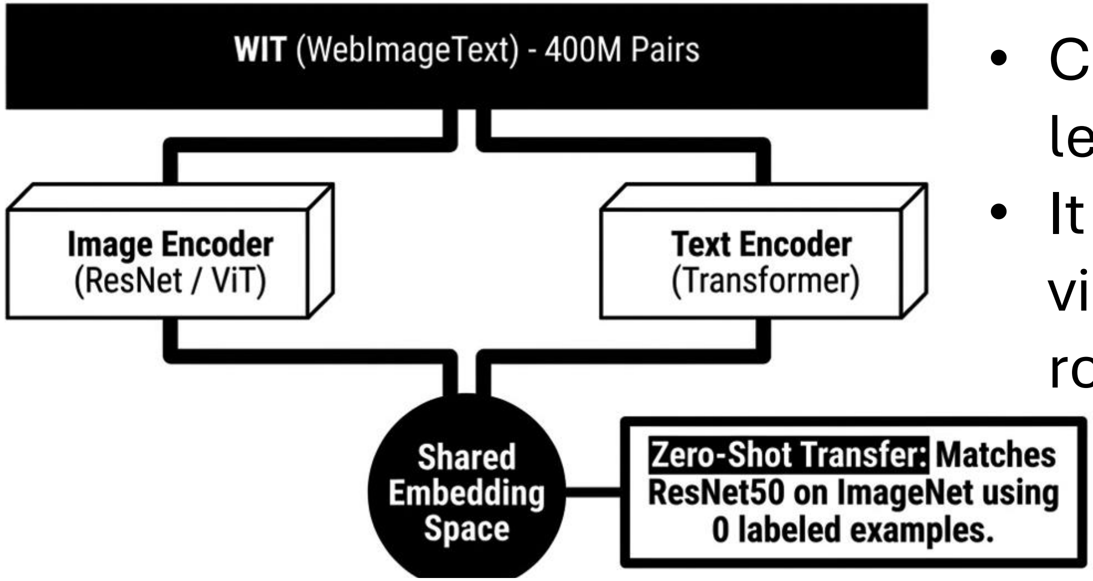


Modality Synthesis: pulling related entitles into a Multi-Modal shared Space

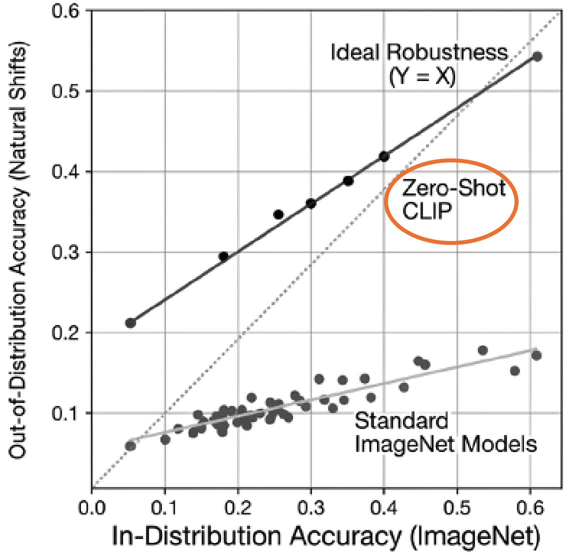


Modalities are no longer isolated. Language becomes a direct, mechanical index for learned visual concepts

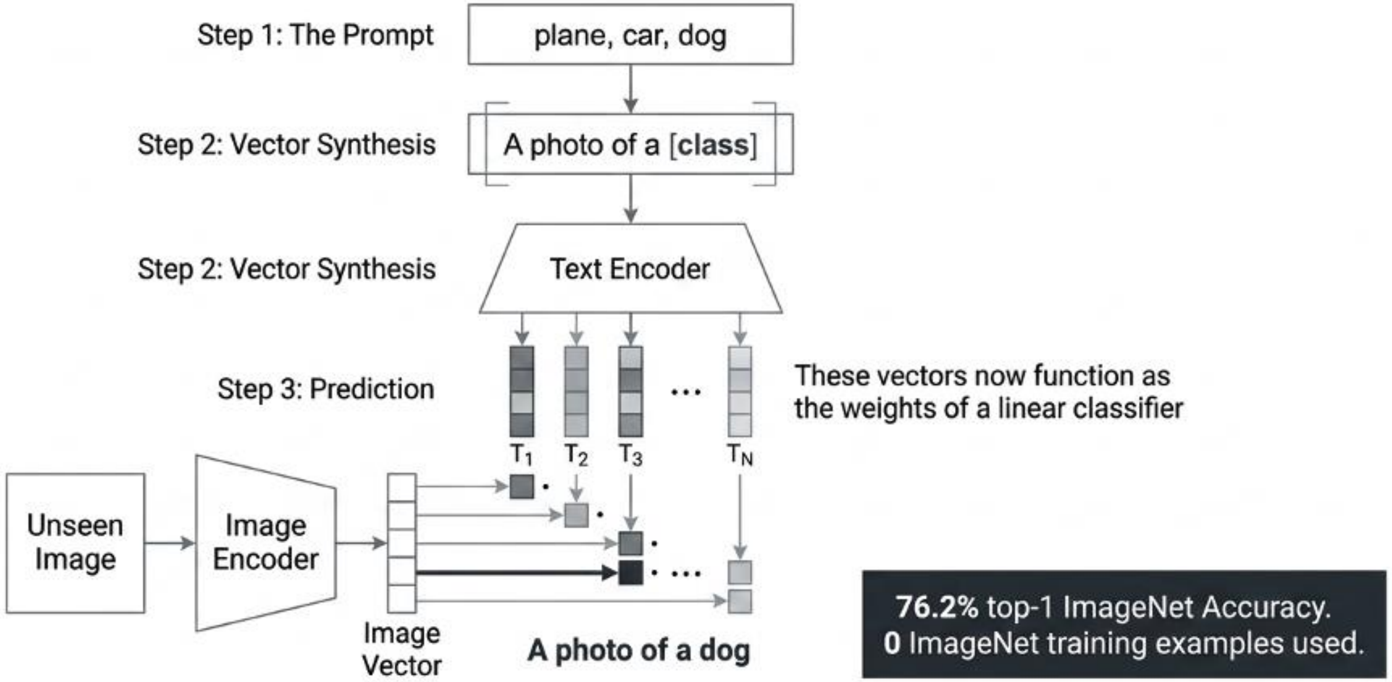
Zero-Shot Inference for Downstream Classification Task



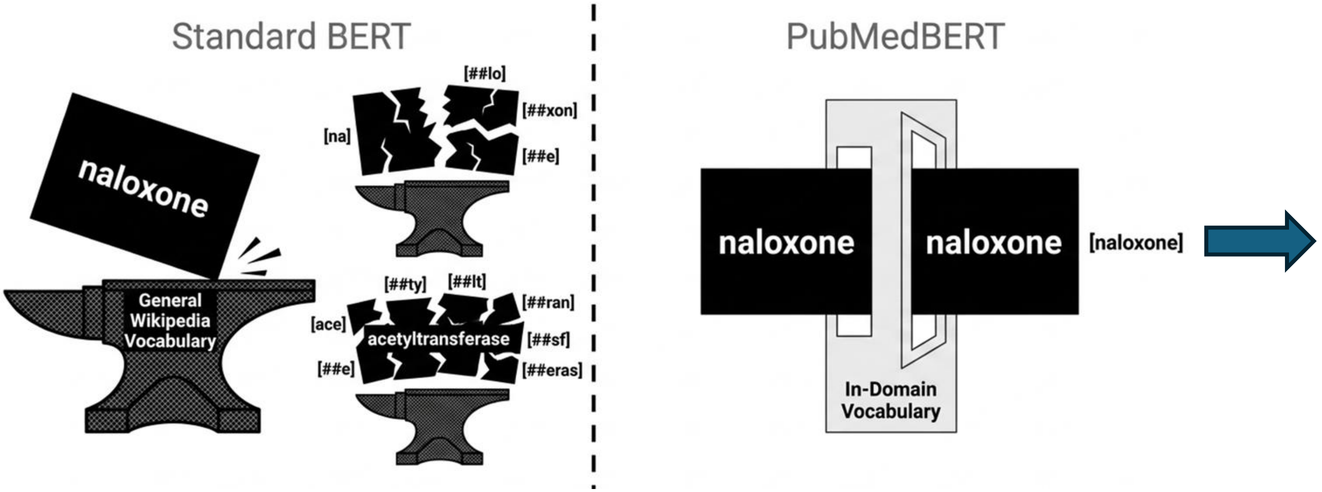
- CLIP broke the reliance on fixed object categories by learning from raw, unlabeled text.
- It proved that natural language supervision can scale visual models to unprecedented, generalized robustness.

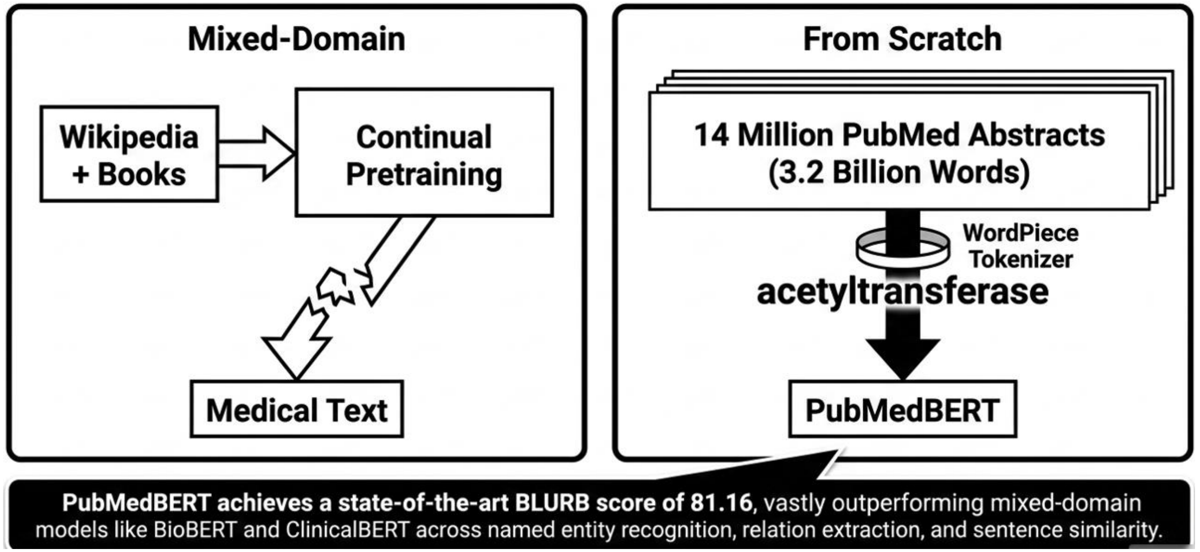


CLIP shrinks the robustness gap by up to 75%



The Gap of Mixed-Domain - From Scratch Paradigm



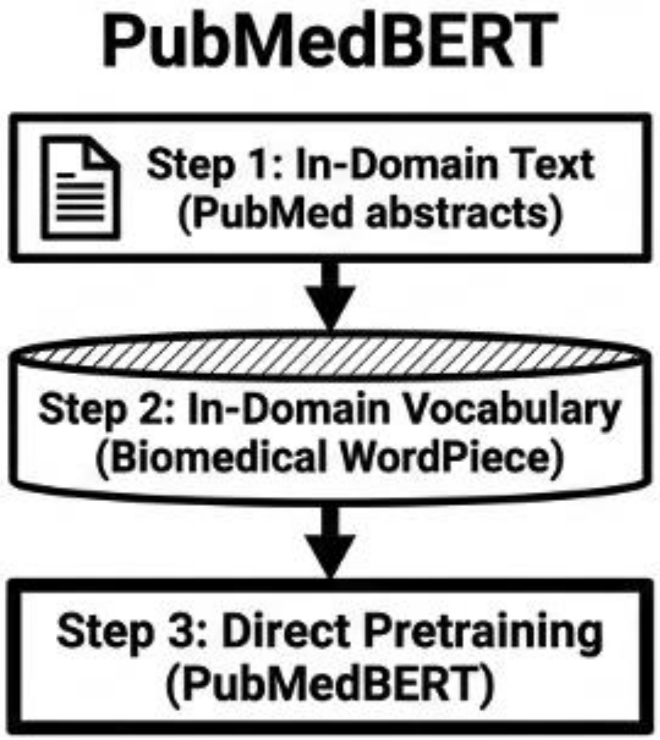


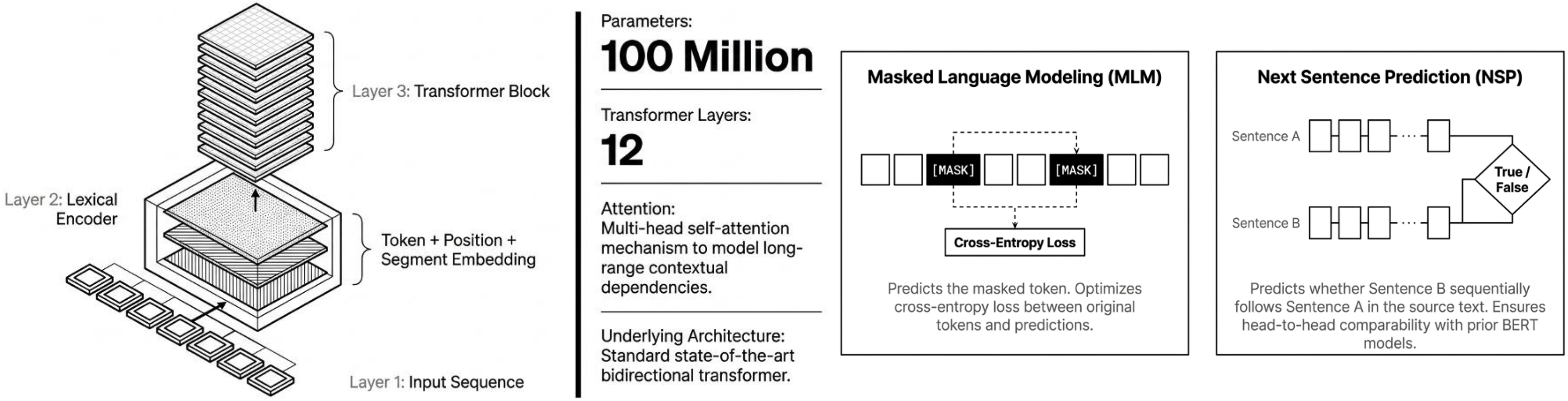
- Internet is not the clinic. Applying general-domain vision-language models to biomedicine triggers catastrophic limitations.
- General tokenizers fail to recognize specialized terminology
- Shatter common medical terms into meaningless fragments

- PubMedBERT use a custom vocabulary and is pretrained only on in-domain text
- Effectively prevent negative transfer from fundamentally different source domains

PubMedBERT Architecture

- New “From Scratch” training pipeline
- BERT-based backbone
- New self-supervised objective (Masked language modeling + Next Sentence Prediction)





Biomedical Language Understanding & Reasoning Benchmark

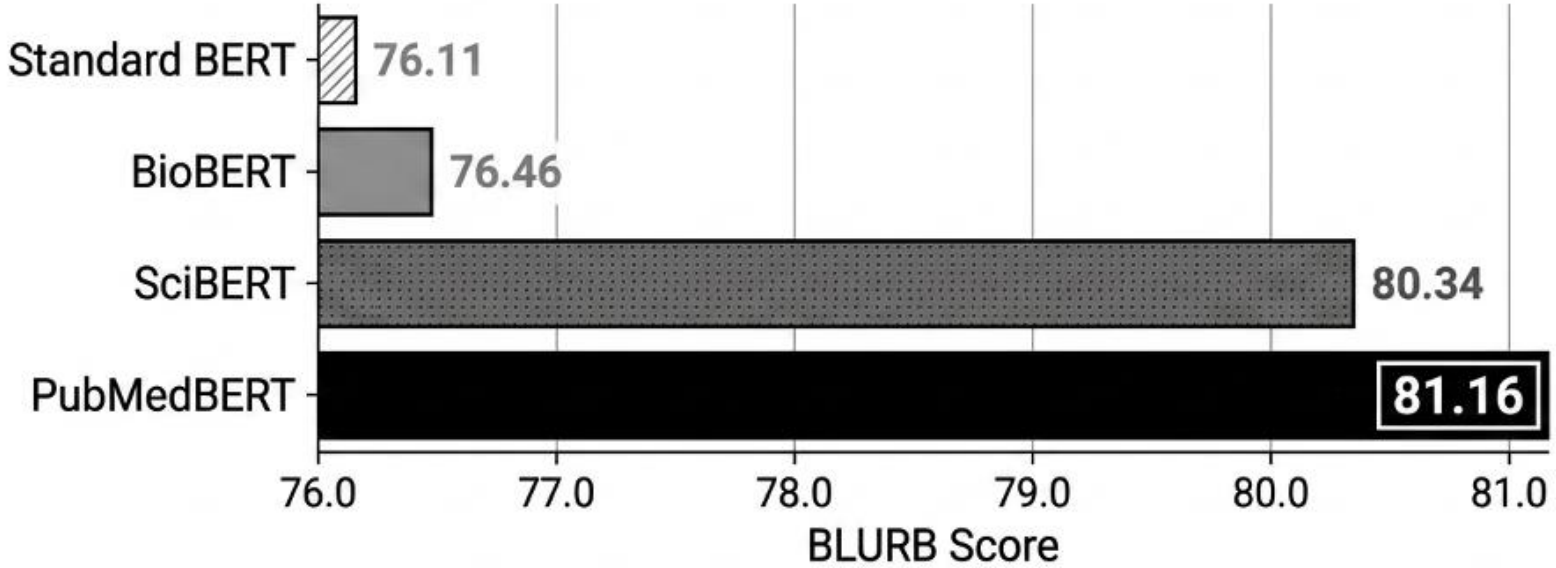
Task Category	Datasets
Named Entity Recognition (NER)	BC5-chem, BC5-disease, NCBI-disease, BC2GM, JNLPBA
Evidence-Based Info (PICO)	EBM PICO
Relation Extraction	ChemProt, DDI, GAD
Sentence Similarity	BIOSSES
Document Classification	HoC
Question Answering	PubMedQA, BioASQ

14 Million
PubMed Abstracts

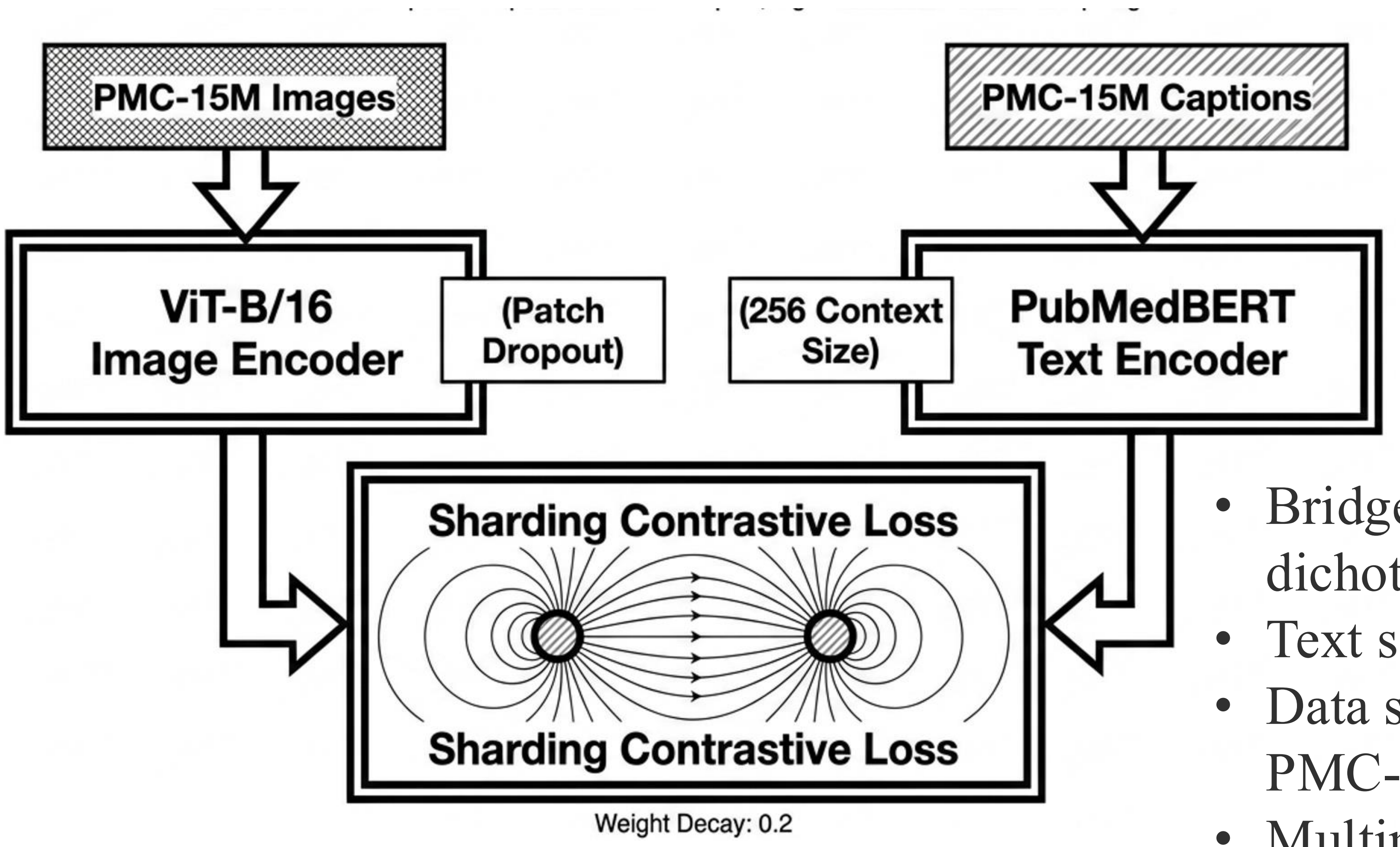
3.2 Billion
Total Words

21 GB
Raw Text Data

Unlike smaller domains that require Wikipedia data to bootstrap, the biomedical domain possesses sufficient mass to train a 100M-parameter model entirely in isolation.

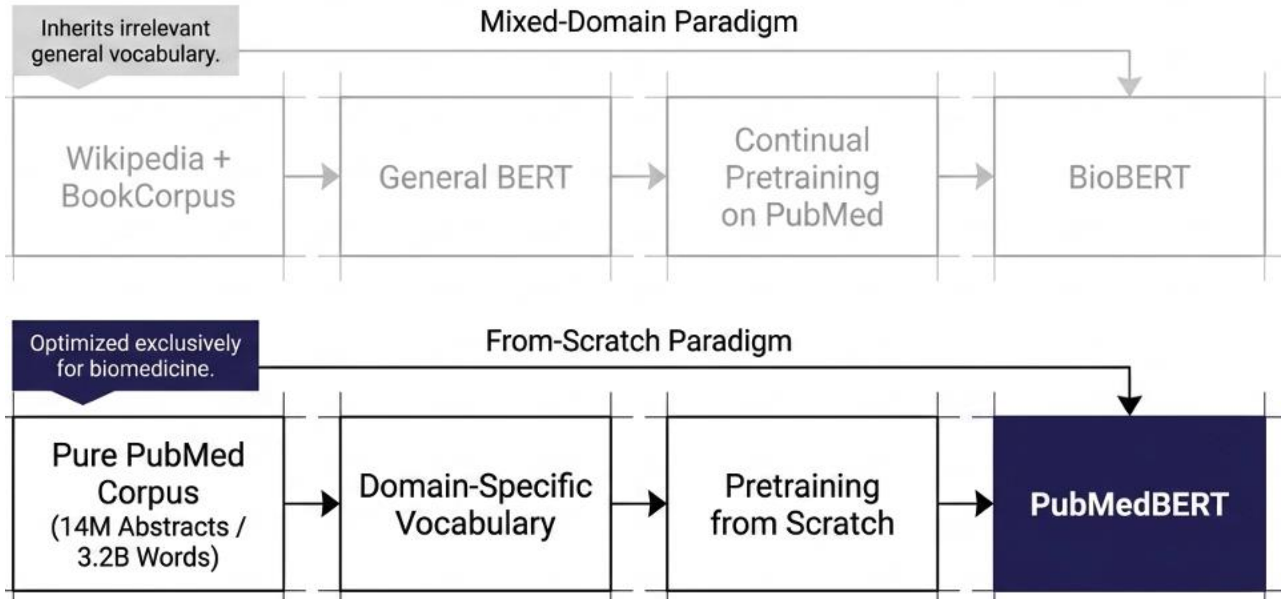


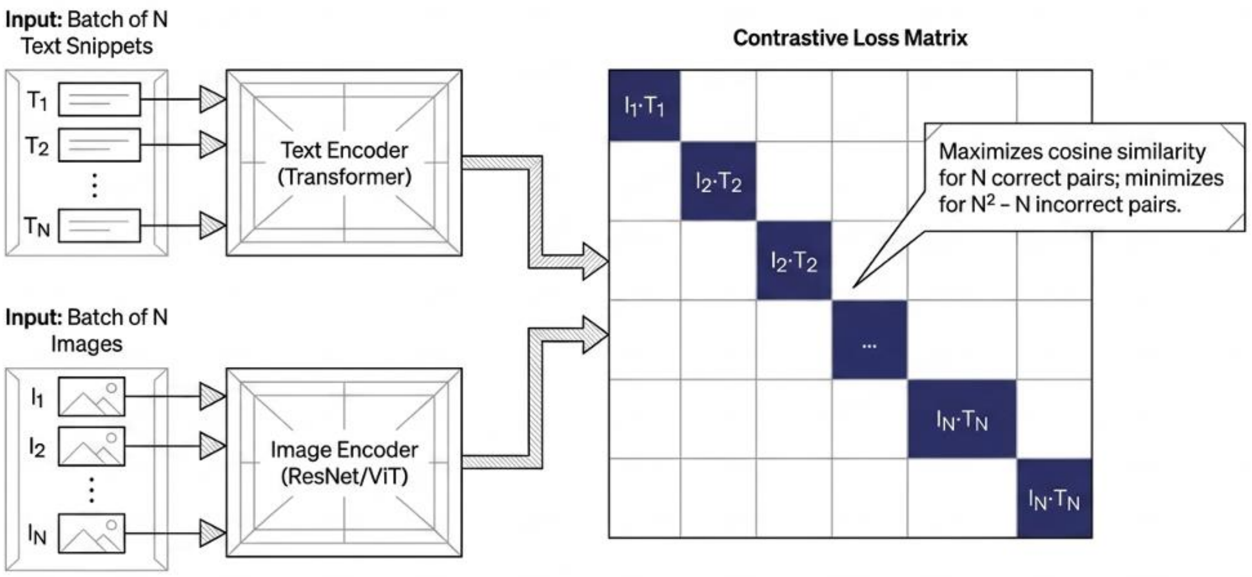
Modality Synthesis – BiomedCLIP Architecture



- Bridges the "biomedical data dichotomy": clinical images and text
- Text side: PubmedBERT integration
- Data side: two orders larger corpus, PMC-15M vs 377.11K MIMIC-CXR
- Multimodal Diversity: 30 major biomedical image types
- Image encoder: ViT-16/B, optimal size, downstream stability, patch dropout

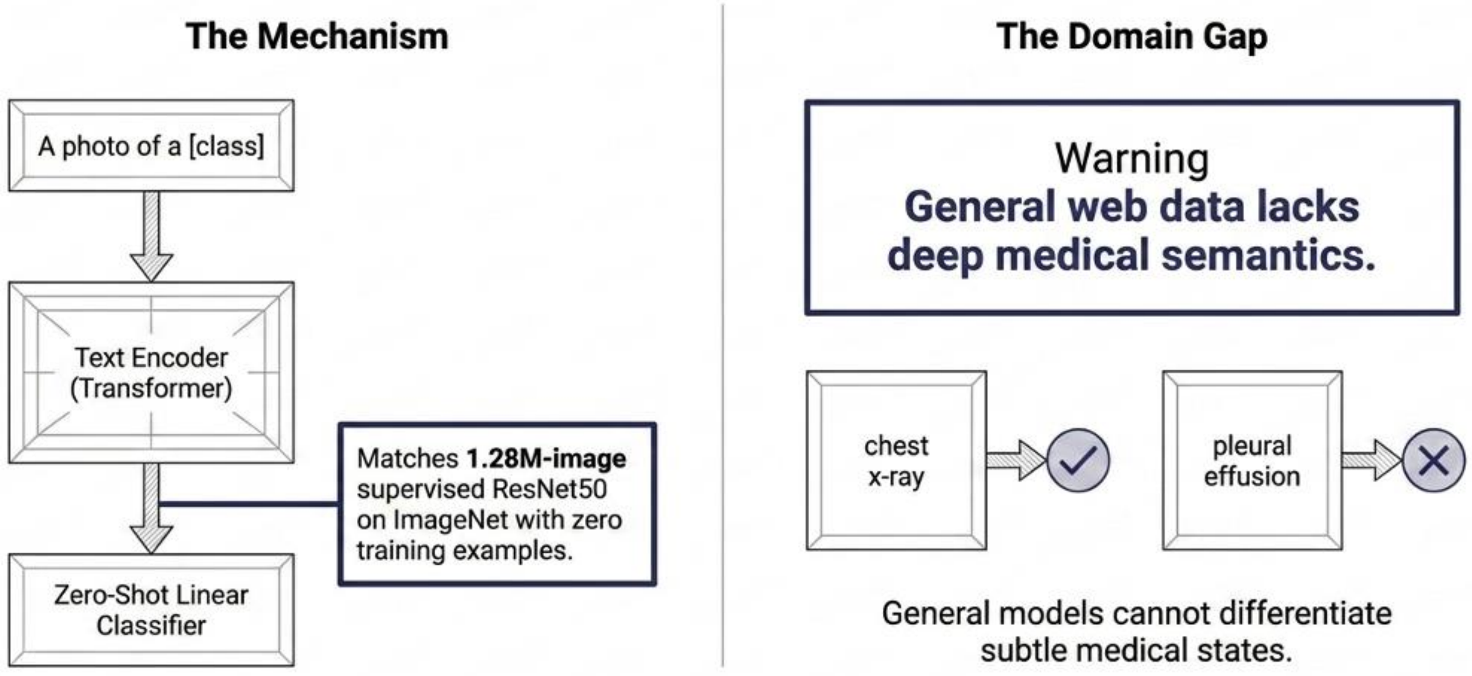
BiomedCLIP Optimization: to Maximize Visual-Language Alignment





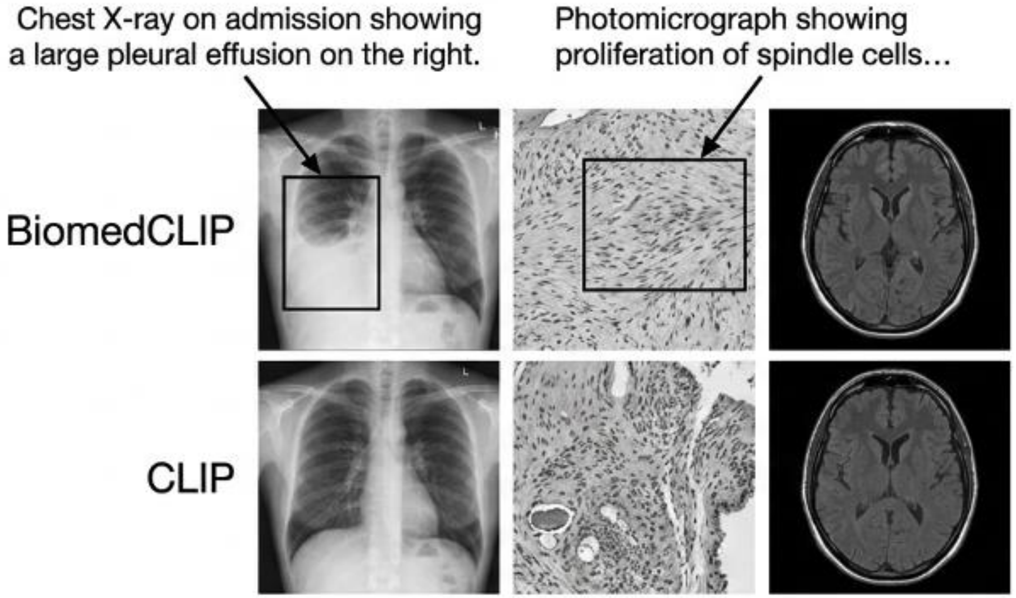
- **Symmetric InfoNCE Loss**: maximizing the cosine similarity between N correct (positive) pairs, while minimizing the similarity for the N2−N incorrect (negative) pairings
- **Directly Optimized Temperature:** optimize a temperature parameter (τ) to controls the scaling of logits within the softmax function
- **Batch Size Tuning: constant batch size of 4k**
- **Overfitting Prevention: weight decay of 0.2**
- **RandomResizedCrop data augmentations**

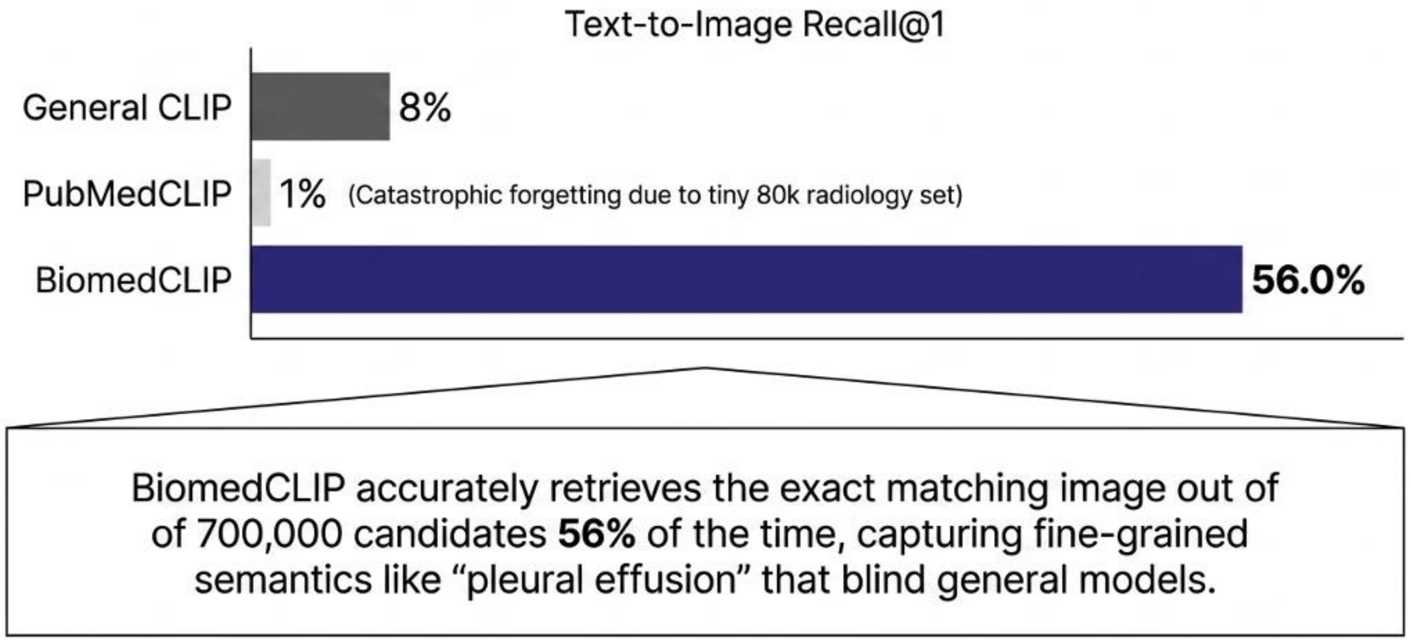
BiomedCLIP Zero-Shot Capability and Bridging Domain Gap

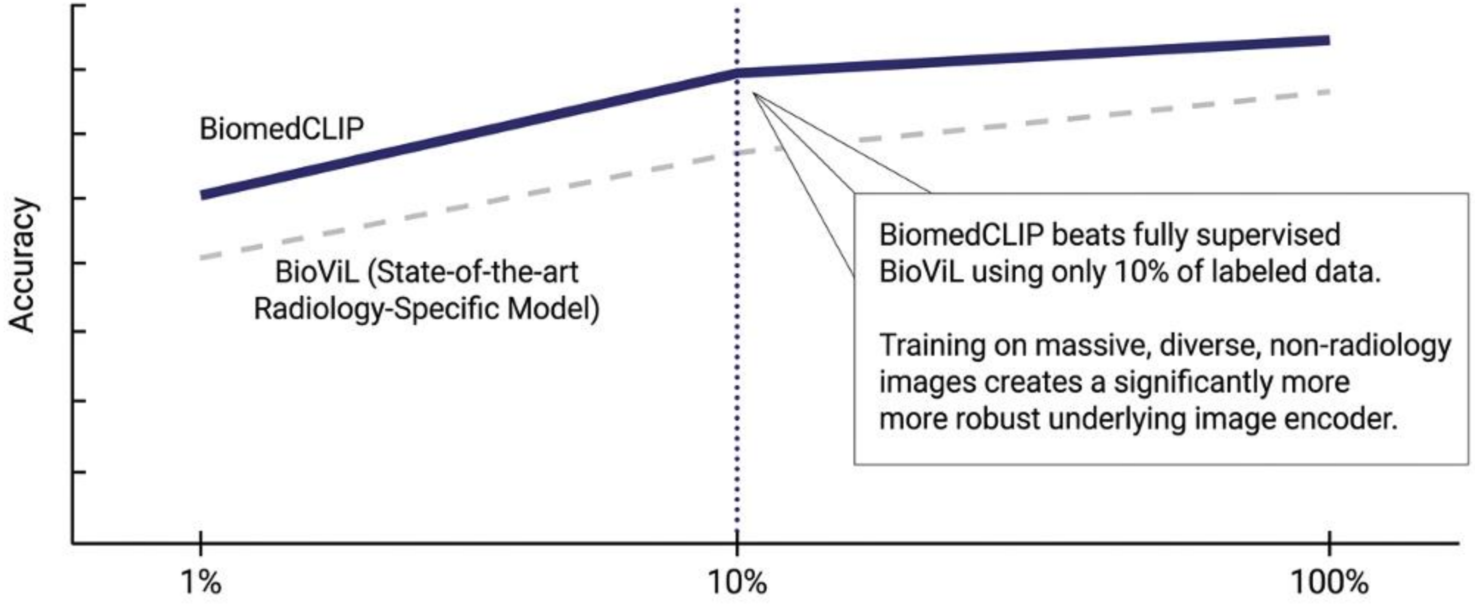


- **Zero-Shot Image Classification:** transform class names into descriptive prompts
- **SOTA Performance**: zero-shot accuracy across major benchmarks – outperforming CLIP, PLIP, PubMedCLIP, excels across diverse sub-specialties
- **Cross-Modal Retrieval**: excel high-level image categories and micro-level caption details

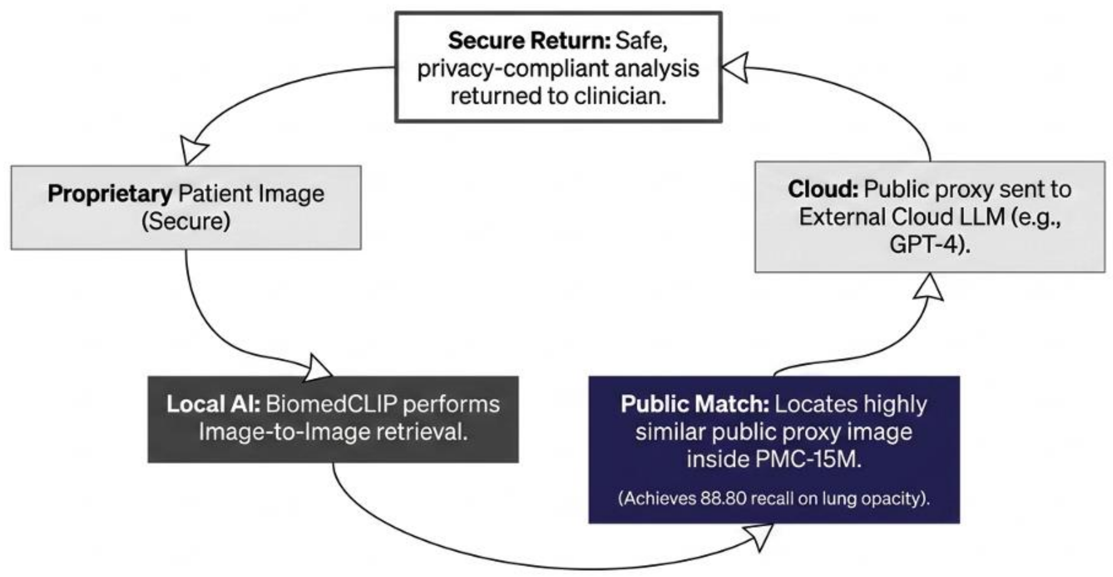
- **Text Length & Context Window** – increase context window from 77 tokens to 256 tokens
- **Medical terminology Shattering** – inherit PubMedBERT's domain-specific WordPiece tokenizer to keep clinical vocabulary
- **Semantic Limitations** – general CLIP models perform badly on biomedical tasks, BiomedCLIP performs better on medical tasks





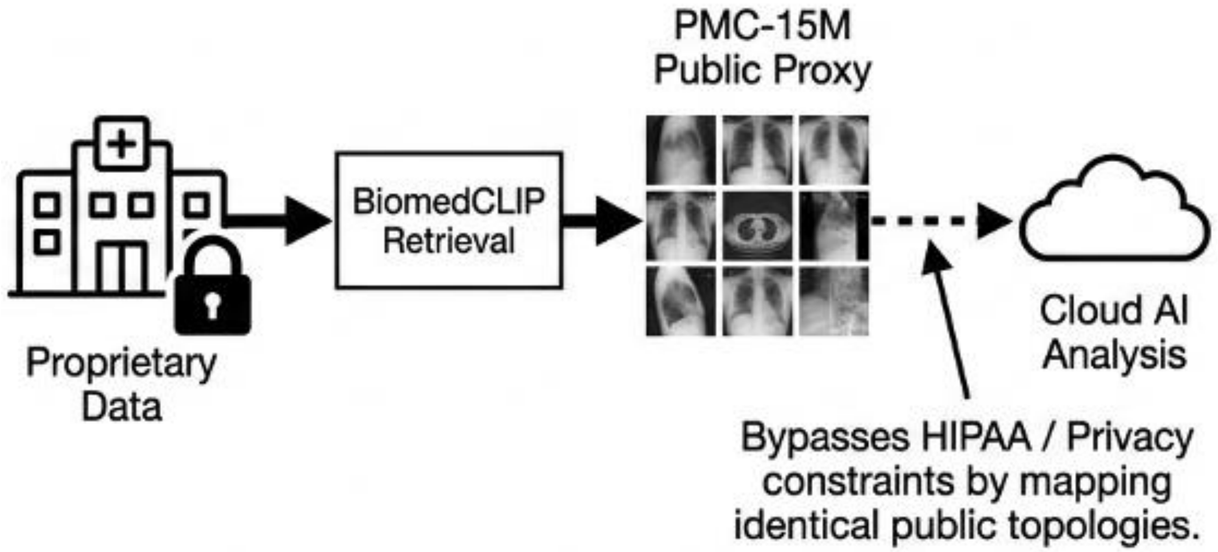


- Retrieves 56% of exact matching image from 700 K candidates
- Effectively captures fine-grained semantics like "pleural effusion"
- Able to map unified biomedical reality rather than memorizing task-specific rules,outperforms task-specific models



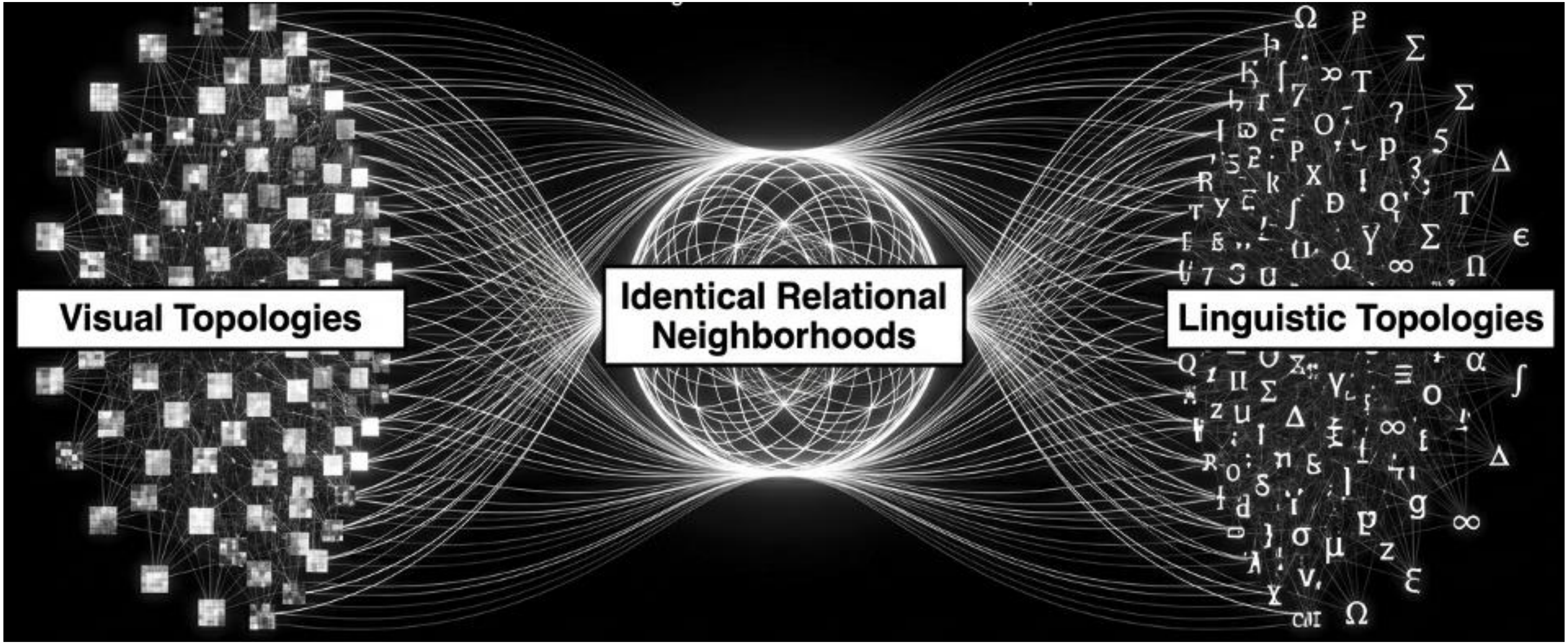
- PMC-15M dataset serves as high-fidelity, privacy-preserving proxies for patient data analysis
- Enable privacy-preserving cloud analysis

- Privacy bottleneck – patient data cannot be shared on the Cloud
- BiomedCLIP as local retriever - process the proprietary image, image-to-image retrieval to search for similar public data
- As a safe "proxy” to link to powerful external cloud models

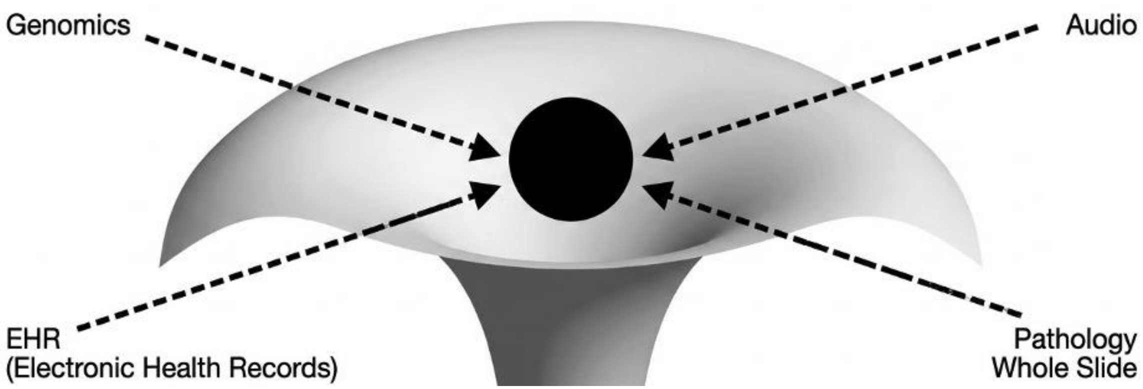


ARH holds: The Topologies Align

BiomedCLIP proves the Aristotelian Representation Hypothesis (ARH). The visual and linguistic encoders are trained to align their internal structures via observing the expansive reality of PMC-15M



Future Projection: A Unified Generalist Space



- ARH dictates that all capable models observing the same reality will eventually align.
- The CLIP style models is not the endpoint, but the blueprint.
- Future medical AI lies in expanding multimodal similarity learning until entirety of mapping all knowledge into a single, unified representation space.

Open questions

- The Future of Foundation Models: Domain-Specific vs. Generalist Scaling
- Integrating Non-Human Modalities under ARH
- The Limits and Ethics of Privacy-Preserving Proxy Analysis
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